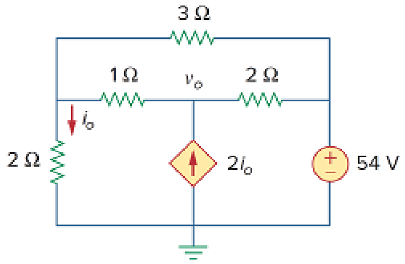
Problem

Find *vo* and *io* in the circuit of Fig. 3.94.

Figure 3.94:



Step-by-step solution

Step 1 of 8

Refer to Figure 3.94 in the text book.

Draw the circuit diagram shown in Figure 1.

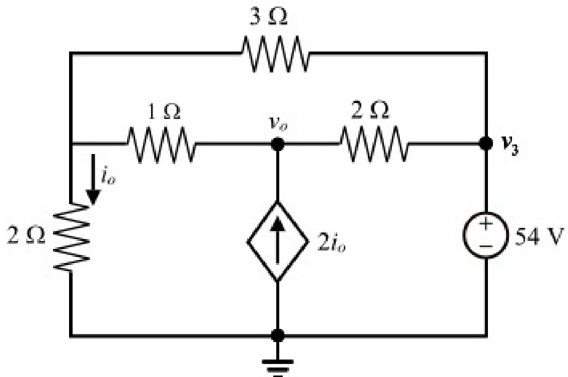


Figure 1

Step 2 of 8

The circuit is labeled and meshes 1 and 2 form a super mesh as shown in Figure 2.

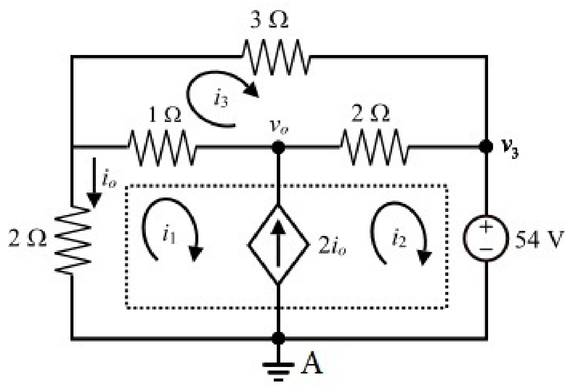


Figure 2

Step 3 of 8

Apply Kirchhoff's voltage law to the super mesh,

$$2i_1 + (i_1 - i_3) + 2(i_2 - i_3) + 54 = 0$$
$$3i_1 + 2i_2 - 3i_3 = -54$$

Therefore $3i_1 + 2i_2 - 3i_3 = -54$ ...... (1)

Apply Kirchhoff's voltage law to mesh 3,

$$3i_3 + 2(i_3 - i_2) + (i_3 - i_1) = 0$$
$$-i_1 - 2i_2 + 6i_3 = 0$$

Therefore $-i_1 - 2i_2 + 6i_3 = 0$ ...... (2)

Apply Kirchhoff's current law at node A,

$$i_2 - i_1 = 2i_o$$

But $i_o = -i_1$

Substitute $-i_1$ for i_o.

$$i_2 - i_1 = -2i_1$$
$$i_2 = -i_1$$

Therefore $i_2 = -i_1$ ...... (3)

Comments (3)

Anonymous

Why does i0 = -i1?

Anonymous

look at the direction of mesh i_1 and compare it to the current i_0.

Anonymous

o_0

Step 4 of 8

Substitute equation (3) in equation (1),

$$3i_1 + 2i_2 - 3i_3 = -54$$
$$3i_1 + 2(-i_1) - 3i_3 = -54$$
$$i_1 - 3i_3 = -54$$

Therefore $i_1 - 3i_3 = -54$ ...... (4)

Substitute equation (3) in equation (2),

$$-i_1 - 2i_2 + 6i_3 = 0$$
$$-i_1 - 2(-i_1) + 6i_3 = 0$$
$$i_1 + 6i_3 = 0$$

Therefore $i_1 + 6i_3 = 0$ ...... (5)

Step 5 of 8

Subtract equation (5) from equation (4),

$$\begin{aligned} i_1 - 3i_3 &= -54 \\ i_1 + 6i_3 &= 0 \\ \hline -9i_3 &= -54 \\ i_3 &= 6\,\text{A} \end{aligned}$$

Substitute $6\,\text{A}$ for i_3 in equation (5),

$$\begin{aligned} i_1 + 6i_3 &= 0 \\ i_1 + 6(6) &= 0 \\ i_1 &= -36\,\text{A} \end{aligned}$$

Substitute $-36\,\text{A}$ for i_1 in equation (3),

$$\begin{aligned} i_2 &= -i_1 \\ &= -(-36) \\ &= 36\,\text{A} \end{aligned}$$

Step 6 of 8

But

$$\begin{aligned} i_o &= -i_1 \\ &= -(-36) \\ &= 36\,\text{A} \end{aligned}$$

Therefore the current i_o is $\boxed{36\,\text{A}}$.

Step 7 of 8

To get v_o, the circuit is shown in Figure 3.

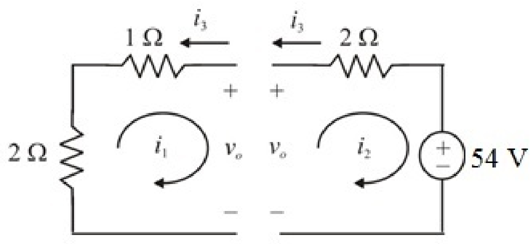


Figure 3

Step 8 of 8

Apply Kirchhoff's voltage law to the loop 1,

$$\begin{aligned} 2i_1 + 1(i_1 - i_3) + v_o &= 0 \\ 3i_1 - 1i_3 + v_o &= 0 \end{aligned}$$

Substitute $6\,\text{A}$ for i_3 and $-36\,\text{A}$ for i_1.

$$\begin{aligned} v_o &= i_3 - 3i_1 \\ &= 6 - 3(-36) \\ &= 6 + 108 \\ &= 114\,\text{V} \end{aligned}$$

Therefore the node voltage v_o is, $\boxed{114\,\text{V}}$.